CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (C.E.)
December 2011
CE – 302 System Programming (S.P.)

Date: 14.12.11, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. Justify the statement: Natural Languages are not Formal Languages while Programming Languages are formal Languages. 3
2. Justify the statement: Assembly Language simplifies programming as compared to Machine Language. 3
3. Define System Programming. Explain the hierarchy of system programming with figure. 3
4. Explain the structure of an Operating System. 2

Q-2

[A] Compare call by value, call by name and call by reference method. 4

[B] Define Job and Scheduling. Explain Job Scheduling in brief. 4

[C] Draw the flow chart of "Program Linking". 4

OR

[A] Write a macro that moves 10 numbers from first operand to the second operand. Take both operands as arguments. 4

[B] Differentiate between Application Software and System Software. Enlist some examples of System software(s). 4

[C] Differentiate between instruction set and addressing modes. 4

Q-3

[A] What is the relationship between system software and the machine structure? Give an example. 4

[B] How does an instruction set affect the assembly language program development? Explain with an example. 4

[C] What is the need for lexical analysis in compiler? What is the role of Lexical Analyzer? 4

OR

Q-3

[A] Write down algorithm for program relocation. 4

[B] What information does symbol table hold? Write about the data structures used to represent the symbol table. Describe about the contents of Symbol Table. 4

[C] What do you mean by parsing? What are the drawbacks of Top Down Parsing? 4

SECTION-II

Q-4

1. What is the use of Program Counter? What is a control register? What is data bus? **3**
2. Explain all data structures required in single pass assembler? **4**
3. What is debugger? What are the methods available for debugging? **4**

Q-5

[A] What do you mean by program relocation? What is known as absolute loader? **4**

[B] What does the object module of a program consist of? What so you mean by self re-locatable program? **4**

[C] What is Editor? What are the functions of an Editor? **4**

OR

Q-5

[A] What is the role of a preprocessor? What are the major functions done by the Compiler? **4**

[B] Explain ENTRY and EXTERN statement with example. Take two assembly language modules M1 and M2. M1 should contain ENTRY statement and M2 should contain EXTERN statement. **4**

[C] Enlist and explain any three types of loading schemes. **4**

Q-6

[A] Explain how the nested macro call is processed? How stack is used to manage the data structures of macros in nested call. **4**

[B] For the given example, apply the algorithm of macro definition processing and fill the necessary tables. **4**

```
              MACRO
              CLEARMEM    &X, &N, &REG=AREG
              LCL         &M
      &M      SET         0
              MOVER       &REG, ='0'
       .MORE  MOVEM       &REG, &X+&M
      &M      SET         &M+1
              AIF         (&M NE N) .MORE
              MEND
```

OR

[B] Discuss about the language features of Java with respect to interpretation. **4**

[C] What is macro preprocessor? How does it work? Justify: Why the number of Actual parameters in the macro call should be less than number of formal parameters in the macro definition? **4**

OR

[C] Discuss about the debugging facilities available in Turbo C++ compiler. **4**